Boids Report

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I. Introduction

This project simulates a flock of animals, based on Craig Reynolds Boids from 1986, using python.

II. Theoretical Background

Craig Reynolds used three steering behaviors, or three rules, when he made his boids in 1986. These are seperation, alignment and cohesion. As explained by Craig Reynolds himself, separation steers boids to avoid crowding local flockmates, alignment steers boids towards the average heading of local flockmates, and cohesion steers boids toward the average position of local flockmates. [1]

III. Design

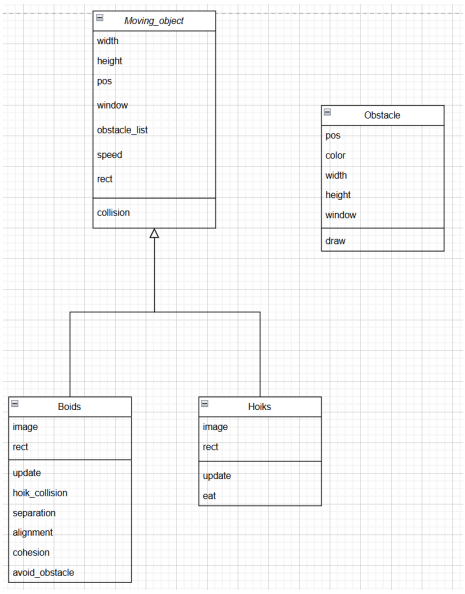


Fig1: Block diagram of the Jetson Xavier NX

The program has four classes: Moving_object, Boids, Hoiks and Obstacle. Boids and Hoiks inherrit from Moving_object because both classes should share most attributes, as well as the collison function.

The Obstacle class has the draw method as its only method. The program has a separate function for creating obstacles.

IV. Implementation

Boids use the three rules separation, alignment and cohesion to behave naturally. Separation keeps boids from crashing into one another, alignment points boids in the same direction as nearby boids and cohesion points boids in the direction of nearby boids. In addition to this the avoid_obstacle method keeps boids from crashing into obstacles. All methods has a factor that is multiplied with the velocity, making it easy to change how important each method is to the boids.

Hoiks has an eat method that finds neargy boids and points the hoik in the direction of the nearest boid. Both the Boids class and the Hoiks class has an update method that updates their position and speed and checks for collisions.

The `create_obstacle()` function creates a rectangular obstacle with random dimentions and a random position.

V. Discussion

There are problems with the code. One of them is that seperation seems to not work properly and boids make little to no effort to avoid other boids. They did previusly avoid eachother, but to an unnatural extent. After making changes to all of the three rules this problem came up and has persisted. The separation factor could theoretically be increased, but this leads to the boids flashing and not moving.

Another problem is that boids seem to loose speed when seperated from other boids and end up standing still after

a while. This could be caused by the alignment method returning -self.speed if no boids are nearby.

The final problem is that boids behave unnatural around obstacles. They move away from obstacles too abruptly and flick from side to side when stuck between two obstacles. A bit of fine tuning should fix both these problems.

VI. Conclusion

In conlusion the boids simulation has a few flaws, but generally works as intended.

References

[1] C. Reynolds, “Boids: Background and Update.” [Online]. Available: http://www.red3d.com/cwr/boids/